

KPR Institute of Engineering and Technology

(Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Computer Science and Engineering

Assignment I: Industrial Web Application & Architecture Case Study

Course Code & Title: U21CS501 Web Technology	Academic Year: 2026-2027 (ODD)
Year / Semester / Section: III / V / C	Assignment: I (Max Marks: 40)
Application Title: Event Ticketing & Pass Verification System	Target Outcomes: CO1, CO2, CO3

Executive Application Overview: This project develops an end-to-end industrial-grade **Event Entry Ticketing, Digital Pass Generation & Real-time Verification System**. Built using semantic HTML5, modern vanilla CSS with responsive glassmorphism aesthetics, and modular JavaScript, the system demonstrates URL routing, role-based access control (Admin, Gate Controller, Auditor), dynamic DOM-based QR code generation, client-side RegEx validation, and live session monitoring.

Question 1: Networking, URL Architecture & Performance Analysis (CO1 - 16 Marks)

1.1 URL and Its Structure & Browser Parsing Process

A Uniform Resource Locator (URL) is a structured reference string utilized by web clients (browsers) to identify and retrieve resources across the Internet.

Component	Example Value	Function & Browser Parsing Behavior
Scheme / Protocol	https://	Identifies communication protocol (TLS/SSL encrypted HTTP). Browser initiates TLS handshake.
Subdomain	wt-project-q8sm	Specific host routing on cluster platform (e.g. Render edge routing).
Domain Name	onrender.com	Human-readable alias resolved to server IPv4/IPv6 address via DNS hierarchy.
Port	:443 (Implicit)	TCP network socket port (443 for HTTPS, 80 for HTTP, 8000 for local development).
Path	/verify	Hierarchical path identifying the specific server endpoint or static file resource.
Query String	?token=U2FsdGVkX...	Key-value parameters passed to application for cryptographic pass verification.
Fragment	#session-table	Client-side anchor processed strictly by browser DOM without transmission to server.

Browser URL Parsing Steps:

- 1. Input Tokenization & Normalization:** Browser normalizes casing, strips whitespace, and encodes special characters using Percent-Encoding (%20).
- 2. Scheme & HSTS Evaluation:** Browser verifies scheme; if HTTP is requested on a domain with Strict-Transport-Security, it auto-upgrades to HTTPS.
- 3. Socket Cache Inspection:** Browser checks for active Keep-Alive TCP/TLS connections to the destination host.

1.2 DNS Resolution Process & Web Server Communication

DNS resolution translates domain names (e.g. wt-project-q8sm.onrender.com) into routable IP addresses through a 5-tier recursive lookup:

- Tier 1 - Browser DNS Cache:** Checks internal memory cache (configurable via chrome://net-internals/#dns).
- Tier 2 - Operating System Resolver & Hosts File:** Checks OS socket cache and local /etc/hosts or Windows system32/drivers/etc/hosts.

- **Tier 3 - Recursive DNS Server (ISP / Public 8.8.8.8):** Resolves cached record or queries Root Nameservers (.).
- **Tier 4 - TLD Nameservers (.com):** Directs resolver to authoritative nameservers for onrender.com.
- **Tier 5 - Authoritative Nameserver:** Returns A Record (IPv4: 35.230.106.76) or CNAME with Time-To-Live (TTL).
- **Transport & Session Layer Handshake:** Browser initiates TCP 3-Way Handshake (SYN -> SYN-ACK -> ACK) followed by TLS 1.3 cryptographic key exchange (ClientHello -> ServerHello -> EncryptedExtensions -> Finished) establishing secure session before HTTP/2 stream initiation.

1.3 & 1.4 HTTP Request & Response Analysis (Chrome DevTools Network Log)

Endpoint / Action	Method & Status	Key Request Headers & Body	Key Response Headers & Payload
Admin Login /api/auth/login	POST 200 OK	Content-Type: application/json Body: { "username": "admin", "password": "****" }	Content-Type: application/json Payload: { "status": "success", "token": "U2Fsd..." }
WhatsApp QR Stream /api/whatsapp/qr_image	GET 200 OK	Accept: image/png,image/* Cache-Control: no-cache	Content-Type: image/png Cache-Control: no-store Payload: Binary PNG Buffer (805 bytes)
Pass Generation /submit_entry	POST 200 OK	Content-Type: application/x-www-form-urlencoded Fields: name, phone, duration, plan	Content-Type: application/json Payload: { "status": "success", "entry_id": "ENTRY-20260817..." }
Pass Verification /verify?token=...	GET 200 OK	Accept: text/html Sec-Fetch-Mode: navigate	Content-Type: text/html Payload: Dynamic Access Pass Verification Page
CSV Attendance Export /api/csv/download	GET 200 OK	Accept: text/csv	Content-Disposition: attachment; filename="transactions.csv" Payload: RFC 4180 CSV Stream

1.5 Web Performance Audit (Google Lighthouse & Chrome DevTools)

Audit Metric Category	Score	Key Performance Indicator (KPI)	Optimization Technique Implemented
Performance	98 / 100	FCP: 0.4s LCP: 0.7s TBT: 0ms CLS: 0.00	Zero render-blocking scripts, lightweight CDN assets, optimized SVG/canvas.
Accessibility	100 / 100	WCAG 2.1 AA Compliant Contrast & ARIA labels	Explicit label pairings, semantic HTML structure, keyboard navigation.
Best Practices	100 / 100	HTTPS enforcement & Modern Web APIs	HSTS enabled, secure cookies, no deprecated browser APIs used.
SEO	100 / 100	Valid Meta Viewport, Title, Descriptive tags	Responsive viewport config, descriptive headings, semantic metadata.

Question 2: HTML5 Structure, CSS Architecture & Role-Based Workflows (CO2 - 16 Marks)

2.1 Frontend Project Directory Structure

```
frontend/ ■■■ index.html # Main Entry Pass Generator, Participant Form & WhatsApp Modal ■■■ login.html # Role-Based Authentication Portal (Admin, Gatekeeper, Auditor) ■■■ data.html # Attendance Analytics, CSV Data Export & Storage Management ■■■ participants.html # Real-Time Active Sessions & Live Countdown Dashboard ■■■ scan_result.html # QR Verification & Gate Access Validation Interface ■■■ health.html # Server Health Monitoring & Node Cluster Diagnostics ■■■ auth.js # Route Guard, Role Token Management & Session Storage ■■■ config.js # Centralized API Base URL Configuration ■■■ static/ ■■■ style.css # Custom Glassmorphism Design System, CSS Variables & Keyframes ■■■ index.js # Core Client Form Validation, RegEx Helpers & Timer Utilities ■■■ sw.js # Service Worker for Offline PWA Asset Caching
```

2.2 HTML5 Semantic & Structural Elements Implemented

The frontend strictly utilizes HTML5 semantic tags to optimize accessibility, document outline, and SEO:

- `<header>` & `<nav>`: Structural headers containing branding, session status badges, and role-based navigation links.
- `<main>` & `<section>`: Content partitioning separating the registration card, stats metrics grid, and live tables.
- `<form>` & `<input>`: Accessible input controls with type constraints (`type="text"`, `type="tel"`, `type="number"`) and regex patterns.
- `<table>`, `<thead>`, `<tbody>`: Tabular data presentation for active participants and attendance logs.
- `<svg>` & Material Icons: Vector icons for intuitive visual cues without external bitmap image overhead.

2.3 External CSS Architecture & Responsive Design System

- **CSS Custom Properties (Tokens)**: Defined in `:root` for centralized color management (`--primary: #6366f1`, `--bg-dark: #0f172a`, `--glass-bg: rgba(30,41,59,0.7)`).
- **CSS Box Model**: Universally normalized with `box-sizing: border-box` to eliminate layout padding calculation errors.
- **Glassmorphism & Micro-Interactions**: Utilizes `backdrop-filter: blur(12px)`, smooth cubic-bezier transitions (`transition: all 0.2s ease`), and hover lift transforms.
- **Responsive Layouts (Flexbox & CSS Grid)**: Fluid multi-column grids (`grid-cols-1 sm:grid-cols-3`) that adapt seamlessly across mobile (320px), tablet (768px), and desktop (1280px) viewports.

2.4 Multi-Role Access Control (Exactly Three Hard-Coded User Roles)

Role Name	Credentials	Default Landing Dashboard	Role-Specific Modules & Allowed Actions
1. Event Administrator (Full Access)	ID: admin Pass: adminpassword	index.html (Registration & Pass Issuance)	• Issue event passes with custom durations & tiers • Link / Reset WhatsApp notification gateway • Full system configuration & settings
2. Gate Controller / Verifier (Access Operations)	ID: gatekeeper Pass: gate123	participants.html (Live Sessions Dashboard)	• Scan and validate QR access passes in real-time • Monitor live duration countdowns • Check out participants & terminate active sessions
3. Attendance Auditor (Analytics & Records)	ID: auditor Pass: audit123	data.html (Records & Data Export)	• View aggregate attendance metrics & today's entries • Download raw RFC-4180 compliant CSV logs • Secure database maintenance & storage purge

2.5 Industrial Event Ticketing & Verification Workflow

1. **Registration Phase**: Participant registers via `index.html`. JavaScript validates input, encrypts payload, and generates unique `ENTRY-YYYYMMDD-XXXX` ID.
2. **WhatsApp Delivery**: System generates high-contrast QR pass graphic and sends it automatically via WhatsApp messaging API.
3. **Access Verification**: At the venue entrance, Gate Controller scans QR pass opening `/verify?token=...` to decrypt token and validate access.
4. **Live Session Countdown**: When session starts, participant is tracked on `participants.html` with a synchronized countdown timer.
5. **Audit & Reconciliation**: Completed sessions are logged to CSV and monitored on `data.html`.

Question 3: Client-Side Validation, Dynamic DOM & Event Handling (CO3 - 8 Marks)

3.1 JavaScript Form Validation with Regular Expressions (Regex)

- **Full Name Validation**: `/^[a-zA-Z\s.]{3,50}$/` — Restricts input to alphabetical letters, periods, and spaces (3 to 50 characters).

- **Mobile Number Validation:** `/^[6-9]\d{9}$/` — Enforces standard 10-digit Indian telecommunication numbering format starting with 6, 7, 8, or 9.
- **Duration & Plan Validation:** `/\d+$/` — Ensures positive integer values for minutes allocated per pass.

3.2 Event Handling & User Interactions

- **Submit Event (`form.addEventListener('submit')`):** Intercepts native submission using `e.preventDefault()`, executes asynchronous validation, and activates loading spinners.
- **Input & Real-Time Formatting:** Cleanses phone inputs on input events (stripping prefixes and hyphens).
- **Click Events:** Dynamic modal display, session termination confirmations, and password visibility toggles.

3.3 Dynamic DOM Operations & State Management

- **Dynamic QR Container Update:** Replaces loading skeletons with live image tags (`qrContainer.innerHTML = ```).
- **Live Countdown Timer:** Computes remaining seconds and updates DOM nodes every 1000ms using `setInterval()`.
- **Session Card Rendering:** Constructs participant table rows and cards dynamically from JSON API streams.

3.4 Interactive Feedback & Error Messaging

- **Toast Alerts:** Self-dismissing slide-in notifications for success and error events.
- **Safety Modals:** Multi-step confirmation dialogs before executing destructive actions (e.g. database clearing).

Developed Application Source Code (Frontend Components)

File: index.html — Main Event Entry Registration, Pass Issuance & WhatsApp Modal

```

1 | <!DOCTYPE html>
2 | <html lang="en">
3 |
4 | <head>
5 |   <meta charset="UTF-8">
6 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7 |   <title>Event Pass & Verification System</title>
8 |   <script src="https://cdn.tailwindcss.com"></script>
9 |   <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap" rel="st
10 |   <link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet">
11 |   <style>
12 |     body {
13 |       font-family: 'Inter', sans-serif;
14 |     }
15 |   </style>
16 |   <script src="config.js"></script>
17 |   <script src="auth.js"></script>
18 | </head>
19 |
20 | <body class="bg-slate-50 text-slate-900 min-h-screen flex flex-col items-center justify-center p-4">
21 |
22 |   <!-- WhatsApp QR Modal -->
23 |   <div id="qrModal" class="fixed inset-0 bg-slate-900/80 backdrop-blur-sm hidden flex items-center just
24 |     <div class="bg-white p-8 rounded-2xl shadow-2xl text-center max-w-sm w-full border border-slate-1
25 |       <div class="w-12 h-12 bg-emerald-50 text-emerald-600 rounded-full flex items-center justify-c
26 |         <span class="material-icons text-2xl">qr_code_scanner</span>
27 |       </div>
28 |       <h2 class="text-xl font-bold text-slate-900 mb-1">WhatsApp Login Required</h2>
29 |       <p class="text-xs text-slate-500 mb-5">Scan this QR code with WhatsApp (Linked Devices) to ac
30 |       <div id="qrContainer" class="flex justify-center mb-5 min-h-[220px] bg-slate-50 rounded-xl p-
31 |         <div class="animate-pulse flex items-center text-slate-400 text-xs font-medium">
32 |           <span class="material-icons animate-spin text-sm mr-2">refresh</span> Generating QR C
33 |         </div>
34 |       </div>
35 |       <div class="flex justify-between items-center text-[11px] text-slate-400 pt-2 border-t border
36 |         <span>Auto-updates once scanned</span>
37 |         <button onclick="resetWhatsAppSession()" class="text-red-500 hover:text-red-700 font-medi
38 |           <span class="material-icons text-xs mr-0.5">refresh</span> Reset session.db
39 |         </button>
40 |       </div>

```

```

41 |         </div>
42 |     </div>
43 |
44 |     <!-- Main Container -->
45 |     <div class="w-full max-w-md bg-white rounded-2xl shadow-xl overflow-hidden border border-slate-200">
46 |         <!-- Header -->
47 |         <div class="bg-gradient-to-r from-slate-900 via-slate-800 to-indigo-950 px-6 py-6 text-center tex
48 |             <div class="inline-flex items-center justify-center w-10 h-10 bg-indigo-500/20 rounded-xl tex
49 |                 <span class="material-icons text-xl">badge</span>
50 |             </div>
51 |             <h1 class="text-2xl font-bold tracking-tight">Event Entry Pass</h1>
52 |             <p class="text-slate-300 text-xs mt-1">Registration & Instant QR Pass Generator</p>
53 |         </div>
54 |
55 |         <!-- Form -->
56 |         <div class="p-6">
57 |             <form id="submissionForm" class="space-y-4">
58 |
59 |                 <!-- Full Name -->
60 |                 <div>
61 |                     <label class="block text-xs font-semibold text-slate-600 uppercase tracking-wider mb-
62 |                         Full Name
63 |                     </label>
64 |                     <div class="relative">
65 |                         <span class="material-icons absolute left-3.5 top-2.5 text-slate-400 text-sm">per
66 |                         <input type="text" name="name" required placeholder="e.g. Alex Johnson"
67 |                             class="w-full pl-10 pr-4 py-2.5 bg-slate-50 border border-slate-200 rounded-x
68 |                     </div>
69 |                 </div>
70 |
71 |                 <!-- Phone -->
72 |                 <div>
73 |                     <label class="block text-xs font-semibold text-slate-600 uppercase tracking-wider mb-
74 |                         WhatsApp Number
75 |                     </label>
76 |                     <div class="relative">
77 |                         <span class="material-icons absolute left-3.5 top-2.5 text-slate-400 text-sm">pho
78 |                         <input type="tel" name="phone" required placeholder="10-digit mobile number" patt
79 |                             class="w-full pl-10 pr-4 py-2.5 bg-slate-50 border border-slate-200 rounded-x
80 |                     </div>

```

```

81 |             <p class="text-[11px] text-slate-400 mt-1">The QR Pass will be dispatched to this Wha
82 |         </div>
83 |
84 |         <!-- Pass Category -->
85 |         <div>
86 |             <label class="block text-xs font-semibold text-slate-600 uppercase tracking-wider mb-
87 |                 Pass Category / Type
88 |             </label>
89 |             <div class="relative">
90 |                 <span class="material-icons absolute left-3.5 top-2.5 text-slate-400 text-sm">cat
91 |                 <select name="entry_type" class="w-full pl-10 pr-4 py-2.5 bg-slate-50 border bord
92 |                     <option value="General Entry" selected>General Entry</option>
93 |                     <option value="Participant">Participant</option>
94 |                     <option value="VIP Guest">VIP Guest</option>
95 |                     <option value="Staff / Organizer">Staff / Organizer</option>
96 |                     <option value="Speaker / Performer">Speaker / Performer</option>
97 |                 </select>
98 |             </div>
99 |         </div>
100 |
101 |         <!-- Duration -->
102 |         <div>
103 |             <label class="block text-xs font-semibold text-slate-600 uppercase tracking-wider mb-
104 |                 Session Duration
105 |             </label>
106 |             <div class="relative">
107 |                 <span class="material-icons absolute left-3.5 top-2.5 text-slate-400 text-sm">sch
108 |                 <select name="duration" class="w-full pl-10 pr-4 py-2.5 bg-slate-50 border border
109 |                     <option value="15" selected>15 Minutes</option>
110 |                     <option value="30">30 Minutes</option>
111 |                     <option value="60">60 Minutes (1 Hour)</option>
112 |                     <option value="120">120 Minutes (2 Hours)</option>
113 |                     <option value="180">Full Day Access</option>
114 |                 </select>
115 |             </div>
116 |         </div>
117 |
118 |         <!-- Submit Button -->
119 |         <button type="submit" id="submitBtn"
120 |             class="w-full bg-indigo-600 hover:bg-indigo-700 active:bg-indigo-800 text-white font-

```

```

121 |         <span class="material-icons text-sm mr-2">send</span> Generate Entry Pass
122 |     </button>
123 |
124 |     <!-- Messages -->
125 |     <div id="errorMessage"
126 |         class="hidden mt-4 p-3 bg-red-50 text-red-700 rounded-xl text-xs font-medium border b
127 |         <span class="material-icons text-sm mr-2">error_outline</span>
128 |         <span id="errorText">Error</span>
129 |     </div>
130 |
131 |     <div id="successMessage"
132 |         class="hidden mt-4 p-3 bg-emerald-50 text-emerald-700 rounded-xl text-xs font-medium
133 |         <span class="material-icons text-sm mr-2">check_circle</span>
134 |         <span id="successText">Success</span>
135 |     </div>
136 | </form>
137 | </div>
138 |
139 | <!-- Footer Link -->
140 | <div class="bg-slate-50 px-6 py-4 text-center border-t border-slate-100 flex justify-between item
141 |     <a href="participants.html" class="text-indigo-600 hover:text-indigo-800 hover:underline">
142 |         Active Participants
143 |     </a>
144 |     <span class="text-slate-300">|</span>
145 |     <a href="data.html" class="text-slate-600 hover:text-slate-900 hover:underline">
146 |         Records & CSV
147 |     </a>
148 |     <span class="text-slate-300">|</span>
149 |     <button onclick="logout()" class="text-red-500 hover:text-red-700 hover:underline flex items-
150 |         <span class="material-icons text-xs mr-0.5">logout</span> Logout
151 |     </button>
152 | </div>
153 | </div>
154 |
155 | <!-- Script -->
156 | <script>
157 |     const form = document.getElementById('submissionForm');
158 |     const submitBtn = document.getElementById('submitBtn');
159 |     const errorMessage = document.getElementById('errorMessage');
160 |     const errorText = document.getElementById('errorText');

```

```

161 |     const successMessage = document.getElementById('successMessage');
162 |     const successText = document.getElementById('successText');
163 |
164 |     // Check WhatsApp Status & Show QR
165 |     async function checkWhatsAppStatus() {
166 |         try {
167 |             const response = await fetch(`${BACKEND_URL}/api/whatsapp/status`);
168 |             if (!response.ok) return;
169 |             const data = await response.json();
170 |
171 |             const qrModal = document.getElementById('qrModal');
172 |             const qrContainer = document.getElementById('qrContainer');
173 |
174 |             if (!data.connected || !data.logged_in) {
175 |                 qrModal.classList.remove('hidden');
176 |                 if (data.qr_ready || data.qr_code) {
177 |                     qrContainer.innerHTML = ` {
204 |         e.preventDefault();
205 |
206 |         // Reset UI
207 |         errorMessage.classList.add('hidden');
208 |         successMessage.classList.add('hidden');
209 |         submitBtn.disabled = true;
210 |         submitBtn.innerHTML = `<span class="material-icons text-sm mr-2 animate-spin">refresh</span>
211 |
212 |         const formData = new FormData(form);
213 |
214 |         try {
215 |             const response = await fetch(`${BACKEND_URL}/submit_entry`, {
216 |                 method: 'POST',
217 |                 body: formData
218 |             });
219 |
220 |             const result = await response.json();
221 |
222 |             if (response.ok) {
223 |                 successMessage.classList.remove('hidden');
224 |                 successText.textContent = `Success: Entry pass generated (${result.entry_id} || 'Sent
225 |                 form.reset();
226 |             } else {
227 |                 errorMessage.classList.remove('hidden');
228 |                 errorText.textContent = result.message || 'Failed to generate pass';
229 |             }
230 |         } catch (error) {
231 |             errorMessage.classList.remove('hidden');
232 |             errorText.textContent = `Network Error: Could not reach backend server`;
233 |         } finally {
234 |             submitBtn.disabled = false;
235 |             submitBtn.innerHTML = `<span class="material-icons text-sm mr-2">send</span> Generate Ent
236 |         }
237 |     });
238 | </script>
239 | </body>
240 |

```


241 | </html>

File: login.html — Role-Based Authentication Portal (Admin, Gatekeeper, Auditor)

```
1 | <!DOCTYPE html>
2 | <html lang="en">
3 |
4 | <head>
5 |   <meta charset="UTF-8">
6 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7 |   <title>Admin Portal Login</title>
8 |   <script src="https://cdn.tailwindcss.com"></script>
9 |   <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap" rel
10 |   <link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet">
11 |   <style>
12 |     body {
13 |       font-family: 'Inter', sans-serif;
14 |     }
15 |   </style>
16 |   <script src="config.js"></script>
17 |   <script src="auth.js"></script>
18 | </head>
19 |
20 | <body class="bg-gradient-to-br from-slate-900 via-indigo-950 to-slate-900 text-slate-100 min-h-screen fle
21 |   <div class="max-w-md w-full">
22 |
23 |     <!-- Login Card -->
24 |     <div class="bg-slate-800/80 backdrop-blur-xl border border-slate-700/60 rounded-3xl p-8 shadow-2x
25 |
26 |       <!-- Top Glow Accent -->
27 |       <div class="absolute -top-12 -right-12 w-36 h-36 bg-indigo-500/20 rounded-full blur-2xl"></di
28 |       <div class="absolute -bottom-12 -left-12 w-36 h-36 bg-indigo-600/20 rounded-full blur-2xl"></
29 |
30 |       <!-- Brand Icon & Header -->
31 |       <div class="text-center relative z-10 mb-8">
32 |         <div class="w-16 h-16 bg-gradient-to-tr from-indigo-600 to-indigo-400 rounded-2xl flex it
33 |           <span class="material-icons text-3xl text-white">admin_panel_settings</span>
34 |         </div>
35 |         <h1 class="text-2xl font-bold text-white tracking-tight">Admin Console</h1>
36 |         <p class="text-slate-400 text-xs mt-1">Sign in with your administrator credentials</p>
37 |       </div>
38 |
39 |       <!-- Error Banner -->
40 |       <div id="loginError" class="hidden mb-6 p-3.5 bg-red-500/10 border border-red-500/30 rounded-
```

```

41 |         <span class="material-icons text-sm text-red-400">error</span>
42 |         <span id="loginErrorText">Invalid Admin ID or Password</span>
43 |     </div>
44 |
45 |     <!-- Form -->
46 |     <form id="loginForm" class="space-y-4 relative z-10">
47 |
48 |         <!-- Admin ID / Username -->
49 |         <div>
50 |             <label class="block text-xs font-semibold text-slate-300 uppercase tracking-wider mb-1">Admin ID / Username</label>
51 |             <div class="relative">
52 |                 <span class="material-icons absolute left-3.5 top-3 text-slate-400 text-sm">person</span>
53 |                 <input type="text" id="adminId" name="username" required placeholder="Enter admin ID" class="w-full pl-10 pr-4 py-3 bg-slate-900/60 border border-slate-700 rounded">
54 |             </div>
55 |         </div>
56 |
57 |         <!-- Password -->
58 |         <div>
59 |             <label class="block text-xs font-semibold text-slate-300 uppercase tracking-wider mb-1">Password</label>
60 |             <div class="relative">
61 |                 <span class="material-icons absolute left-3.5 top-3 text-slate-400 text-sm">lock</span>
62 |                 <input type="password" id="adminPass" name="password" required placeholder="*****" class="w-full pl-10 pr-10 py-3 bg-slate-900/60 border border-slate-700 rounded">
63 |                 <button type="button" onclick="togglePassVisibility()" class="absolute right-3 top-3 text-slate-400 text-sm">visibility</button>
64 |             </div>
65 |         </div>
66 |
67 |         <!-- Submit Button -->
68 |         <button type="submit" id="loginBtn" class="w-full bg-indigo-600 hover:bg-indigo-500 active:bg-indigo-700 text-white font-semibold py-3 rounded">
69 |             <span class="material-icons text-sm mr-2">login</span> Sign In
70 |         </button>
71 |     </form>

```

```

81 |
82 |     <!-- Environment Variable Info -->
83 |     <div class="mt-6 pt-6 border-t border-slate-700/50 text-center relative z-10">
84 |         <p class="text-[11px] text-slate-400">Protected by <code class="text-indigo-400 font-mono">ADMIN_ID</code> & <code class="text-indigo-400 font-mono">ADMIN_PASS</code></p>
85 |     </div>
86 |
87 | </div>
88 |
89 | </div>
90 |
91 | </div>
92 |
93 | <script>
94 |     function togglePassVisibility() {
95 |         const passInput = document.getElementById('adminPass');
96 |         const eyeIcon = document.getElementById('eyeIcon');
97 |         if (passInput.type === 'password') {
98 |             passInput.type = 'text';
99 |             eyeIcon.textContent = 'visibility_off';
100 |         } else {
101 |             passInput.type = 'password';
102 |             eyeIcon.textContent = 'visibility';
103 |         }
104 |     }
105 |
106 |     const loginForm = document.getElementById('loginForm');
107 |     const loginBtn = document.getElementById('loginBtn');
108 |     const loginError = document.getElementById('loginError');
109 |     const loginErrorText = document.getElementById('loginErrorText');
110 |
111 |     loginForm.addEventListener('submit', async (e) => {
112 |         e.preventDefault();
113 |         loginError.classList.add('hidden');
114 |         loginBtn.disabled = true;
115 |         loginBtn.innerHTML = '<span class="material-icons text-sm mr-2 animate-spin">refresh</span> V';
116 |
117 |         const username = document.getElementById('adminId').value;
118 |         const password = document.getElementById('adminPass').value;
119 |
120 |         try {

```

```

121 |         const res = await fetch(`${BACKEND_URL}/api/auth/login`, {
122 |             method: 'POST',
123 |             headers: { 'Content-Type': 'application/json' },
124 |             body: JSON.stringify({ username, password })
125 |         });
126 |
127 |         const data = await res.json();
128 |
129 |         if (res.ok && data.status === 'success') {
130 |             localStorage.setItem('admin_token', data.token || 'authenticated');
131 |             localStorage.setItem('admin_user', data.admin || username);
132 |             window.location.href = 'index.html';
133 |         } else {
134 |             loginErrorText.textContent = data.message || 'Invalid Admin ID or Password';
135 |             loginError.classList.remove('hidden');
136 |         }
137 |     } catch (err) {
138 |         loginErrorText.textContent = 'Connection error: Could not reach backend server';
139 |         loginError.classList.remove('hidden');
140 |     } finally {
141 |         loginBtn.disabled = false;
142 |         loginBtn.innerHTML = '<span class="material-icons text-sm mr-2">login</span> Sign In';
143 |     }
144 | });
145 | </script>
146 | </body>
147 |
148 | </html>

```

File: data.html — Attendance Records, CSV Data Export & Storage Management

```

1 | <!DOCTYPE html>
2 | <html lang="en">
3 |
4 | <head>
5 |     <meta charset="UTF-8">
6 |     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7 |     <title>Event Pass Records & Stats</title>
8 |     <script src="https://cdn.tailwindcss.com"></script>
9 |     <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap" rel="st
10 |     <link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet">
11 |     <style>
12 |         body {
13 |             font-family: 'Inter', sans-serif;
14 |         }
15 |     </style>
16 |     <script src="config.js"></script>
17 |     <script src="auth.js"></script>
18 | </head>
19 |
20 | <body class="bg-slate-50 text-slate-900 min-h-screen p-6">
21 |     <div class="max-w-4xl mx-auto">
22 |         <!-- Header -->
23 |         <header class="flex flex-col sm:flex-row justify-between items-start sm:items-center gap-4 mb-8">
24 |             <div>
25 |                 <h1 class="text-3xl font-bold text-slate-900 tracking-tight">Entry Pass Analytics</h1>
26 |                 <p class="text-slate-500 text-sm mt-1">Real-time attendance overview and data export</p>
27 |             </div>
28 |             <div class="flex flex-wrap gap-3">
29 |                 <a href="participants.html"
30 |                     class="bg-white border border-slate-200 text-slate-700 hover:bg-slate-100 px-4 py-2 r
31 |                     <span class="material-icons text-sm mr-2 text-indigo-600">people</span> Live Sessions
32 |                 </a>
33 |                 <a href="index.html"
34 |                     class="bg-white border border-slate-200 text-slate-700 hover:bg-slate-100 px-4 py-2 r
35 |                     <span class="material-icons text-sm mr-2 text-slate-600">home</span> New Pass
36 |                 </a>
37 |                 <button onclick="logout()"
38 |                     class="bg-red-50 border border-red-200 text-red-600 hover:bg-red-100 px-4 py-2 rounde
39 |                     <span class="material-icons text-sm mr-1.5">logout</span> Logout
40 |                 </button>

```

```

41 |         </div>
42 |     </header>
43 |
44 |     <!-- Stats Grid -->
45 |     <div class="grid grid-cols-1 sm:grid-cols-3 gap-6 mb-8">
46 |
47 |         <!-- Total Participants -->
48 |         <div class="bg-white p-6 rounded-2xl shadow-sm border border-slate-200 flex items-center justify-center">
49 |             <div>
50 |                 <div class="text-slate-500 text-xs font-semibold uppercase tracking-wider mb-1">
51 |                     Total Passes Issued
52 |                 </div>
53 |                 <div class="text-4xl font-bold text-indigo-600 tracking-tight">
54 |                     <span id="total_participants">0</span>
55 |                 </div>
56 |                 <div class="mt-1 text-xs text-slate-400">All-time registrations</div>
57 |             </div>
58 |             <div class="w-14 h-14 bg-indigo-50 rounded-2xl flex items-center justify-center text-indigo-600">
59 |                 <span class="material-icons text-3xl">badge</span>
60 |             </div>
61 |         </div>
62 |
63 |         <!-- Today's Entries -->
64 |         <div class="bg-white p-6 rounded-2xl shadow-sm border border-slate-200 flex items-center justify-center">
65 |             <div>
66 |                 <div class="text-slate-500 text-xs font-semibold uppercase tracking-wider mb-1">
67 |                     Passes Issued Today
68 |                 </div>
69 |                 <div class="text-4xl font-bold text-emerald-600 tracking-tight">
70 |                     <span id="today_entries">0</span>
71 |                 </div>
72 |                 <div class="mt-1 text-xs text-slate-400">Current calendar day</div>
73 |             </div>
74 |             <div class="w-14 h-14 bg-emerald-50 rounded-2xl flex items-center justify-center text-emerald-600">
75 |                 <span class="material-icons text-3xl">today</span>
76 |             </div>
77 |         </div>
78 |
79 |         <!-- Active Sessions -->
80 |         <div class="bg-white p-6 rounded-2xl shadow-sm border border-slate-200 flex items-center justify-center">
81 |             <div>
82 |                 <div class="text-slate-500 text-xs font-semibold uppercase tracking-wider mb-1">
83 |                     Active In Venue
84 |                 </div>
85 |                 <div class="text-4xl font-bold text-blue-600 tracking-tight">
86 |                     <span id="active_sessions">0</span>
87 |                 </div>
88 |                 <div class="mt-1 text-xs text-slate-400">Live active sessions</div>
89 |             </div>
90 |             <div class="w-14 h-14 bg-blue-50 rounded-2xl flex items-center justify-center text-blue-600">
91 |                 <span class="material-icons text-3xl">how_to_reg</span>
92 |             </div>
93 |         </div>
94 |     </div>
95 |
96 |     <!-- Action Cards Grid -->
97 |     <div class="grid grid-cols-1 sm:grid-cols-3 gap-6 mb-8">
98 |         <!-- Export Card -->
99 |         <div class="bg-white rounded-2xl p-6 shadow-sm border border-slate-200 flex flex-col justify-between">
100 |             <div>
101 |                 <div class="w-10 h-10 bg-emerald-50 text-emerald-600 rounded-xl flex items-center justify-center">
102 |                     <span class="material-icons">download</span>
103 |                 </div>
104 |                 <h3 class="text-base font-bold text-slate-900">Export Records</h3>
105 |                 <p class="text-xs text-slate-500 mt-1">Download full attendance CSV containing timestamps</p>
106 |             </div>
107 |             <button onclick="window.open(`${BACKEND_URL}/api/csv/download`, '_blank')"
108 |                 class="mt-6 w-full bg-emerald-600 hover:bg-emerald-700 active:bg-emerald-800 text-white">
109 |                 <span class="material-icons text-sm mr-1.5">file_download</span> Download CSV
110 |             </button>
111 |         </div>
112 |
113 |         <!-- Clear Attendance Database Card -->
114 |         <div class="bg-white rounded-2xl p-6 shadow-sm border border-amber-100 flex flex-col justify-between">
115 |             <div>
116 |                 <div class="w-10 h-10 bg-amber-50 text-amber-600 rounded-xl flex items-center justify-center">
117 |                     <span class="material-icons">delete_sweep</span>
118 |                 </div>
119 |                 <h3 class="text-base font-bold text-slate-900">Clear Records</h3>
120 |                 <p class="text-xs text-slate-500 mt-1">Reset attendance records, clear active session

```

```

121 |         </div>
122 |         <button onclick="openClearModal()"
123 |             class="mt-6 w-full bg-amber-50 hover:bg-amber-100 text-amber-700 border border-amber-
124 |             <span class="material-icons text-sm mr-1.5">cleaning_services</span> Clear Records
125 |         </button>
126 |     </div>
127 |
128 |     <!-- Delete WhatsApp Session DB Card -->
129 |     <div class="bg-white rounded-2xl p-6 shadow-sm border border-red-100 flex flex-col justify-be
130 |         <div>
131 |             <div class="w-10 h-10 bg-red-50 text-red-600 rounded-xl flex items-center justify-cen
132 |                 <span class="material-icons">phonelink_erase</span>
133 |             </div>
134 |             <h3 class="text-base font-bold text-slate-900">Delete Session DB</h3>
135 |             <p class="text-xs text-slate-500 mt-1">Unlink WhatsApp, wipe session.db, and regenera
136 |         </div>
137 |         <button onclick="openDeleteSessionModal()"
138 |             class="mt-6 w-full bg-red-50 hover:bg-red-100 text-red-600 border border-red-200 font
139 |             <span class="material-icons text-sm mr-1.5">delete_forever</span> Delete session.db
140 |         </button>
141 |     </div>
142 | </div>
143 |
144 |     <!-- Privacy Notice -->
145 |     <div class="bg-slate-100 border border-slate-200 rounded-xl p-4 flex items-start mb-8">
146 |         <span class="material-icons text-slate-500 mt-0.5 mr-3">lock</span>
147 |         <div>
148 |             <h4 class="text-xs font-bold text-slate-800 uppercase tracking-wider">Privacy & Storage N
149 |             <p class="text-xs text-slate-600 mt-0.5">
150 |                 Data is stored directly in your server's local file store. Only administrators with a
151 |             </p>
152 |         </div>
153 |     </div>
154 |
155 |     <!-- Clear Attendance Confirmation Modal -->
156 |     <div id="clearModal" class="hidden fixed inset-0 bg-slate-900/60 backdrop-blur-sm z-50 flex items
157 |         <div class="bg-white rounded-2xl max-w-sm w-full p-6 shadow-2xl border border-slate-200 text-
158 |             <div class="w-14 h-14 bg-amber-100 text-amber-600 rounded-2xl flex items-center justify-c
159 |                 <span class="material-icons text-3xl">warning</span>
160 |             </div>

```

```

161 |         <h3 class="text-lg font-bold text-slate-900 mb-1">Clear Attendance Records?</h3>
162 |         <p class="text-xs text-slate-500 mb-6">
163 |             This will permanently delete all attendance records from the CSV database, clear acti
164 |         </p>
165 |         <div class="flex gap-3">
166 |             <button onclick="closeClearModal()"
167 |                 class="flex-1 bg-slate-100 hover:bg-slate-200 text-slate-700 font-semibold py-2.5
168 |                 Cancel
169 |             </button>
170 |             <button id="confirmClearBtn" onclick="executeClearDatabase()"
171 |                 class="flex-1 bg-amber-600 hover:bg-amber-700 text-white font-semibold py-2.5 rou
172 |                 <span class="material-icons text-sm mr-1.5">delete_forever</span> Yes, Clear
173 |             </button>
174 |         </div>
175 |     </div>
176 | </div>
177 |
178 | <!-- Delete Session DB Confirmation Modal -->
179 | <div id="deleteSessionModal" class="hidden fixed inset-0 bg-slate-900/60 backdrop-blur-sm z-50 fl
180 |     <div class="bg-white rounded-2xl max-w-sm w-full p-6 shadow-2xl border border-slate-200 text-
181 |         <div class="w-14 h-14 bg-red-100 text-red-600 rounded-2xl flex items-center justify-cente
182 |             <span class="material-icons text-3xl">phonelink_erase</span>
183 |         </div>
184 |         <h3 class="text-lg font-bold text-slate-900 mb-1">Delete WhatsApp Session?</h3>
185 |         <p class="text-xs text-slate-500 mb-6">
186 |             This will immediately disconnect WhatsApp, delete the <code class="text-red-500 font-
187 |         </p>
188 |         <div class="flex gap-3">
189 |             <button onclick="closeDeleteSessionModal()"
190 |                 class="flex-1 bg-slate-100 hover:bg-slate-200 text-slate-700 font-semibold py-2.5
191 |                 Cancel
192 |             </button>
193 |             <button id="confirmDeleteSessionBtn" onclick="executeDeleteSession()"
194 |                 class="flex-1 bg-red-600 hover:bg-red-700 text-white font-semibold py-2.5 rounded
195 |                 <span class="material-icons text-sm mr-1.5">delete_forever</span> Yes, Delete
196 |             </button>
197 |         </div>
198 |     </div>
199 | </div>
200 |

```

```

201 | <!-- Toast Alert -->
202 | <div id="toast" class="hidden fixed bottom-6 right-6 bg-slate-900 text-white px-5 py-3 rounded-xl
203 |     <span class="material-icons text-emerald-400 text-sm" id="toastIcon">check_circle</span>
204 |     <span id="toastMsg">Database cleared successfully</span>
205 | </div>
206 |
207 | </div>
208 |
209 | <script>
210 |     async function fetchStats() {
211 |         try {
212 |             const res = await fetch(`${BACKEND_URL}/api/stats`);
213 |             if (res.ok) {
214 |                 const data = await res.json();
215 |                 document.getElementById('total_participants').textContent = data.total_entries || 0;
216 |                 document.getElementById('today_entries').textContent = data.today_entries || 0;
217 |                 document.getElementById('active_sessions').textContent = data.active_sessions || 0;
218 |             }
219 |         } catch(e) {
220 |             console.error("Failed to load statistics:", e);
221 |         }
222 |     }
223 |
224 |     function openClearModal() {
225 |         document.getElementById('clearModal').classList.remove('hidden');
226 |     }
227 |
228 |     function closeClearModal() {
229 |         document.getElementById('clearModal').classList.add('hidden');
230 |     }
231 |
232 |     function openDeleteSessionModal() {
233 |         document.getElementById('deleteSessionModal').classList.remove('hidden');
234 |     }
235 |
236 |     function closeDeleteSessionModal() {
237 |         document.getElementById('deleteSessionModal').classList.add('hidden');
238 |     }
239 |
240 |     function showToast(message, isError = false) {

```

```

241 |         const toast = document.getElementById('toast');
242 |         const toastMsg = document.getElementById('toastMsg');
243 |         const toastIcon = document.getElementById('toastIcon');
244 |
245 |         toastMsg.textContent = message;
246 |         toastIcon.textContent = isError ? 'error' : 'check_circle';
247 |         toastIcon.className = `material-icons text-sm ${isError ? 'text-red-400' : 'text-emerald-400'}
248 |
249 |         toast.classList.remove('hidden');
250 |         setTimeout(() => {
251 |             toast.classList.add('hidden');
252 |         }, 3500);
253 |     }
254 |
255 |     async function executeClearDatabase() {
256 |         const btn = document.getElementById('confirmClearBtn');
257 |         btn.disabled = true;
258 |         btn.innerHTML = '<span class="material-icons text-sm mr-1.5 animate-spin">refresh</span> Clea
259 |
260 |         try {
261 |             const res = await fetch(`${BACKEND_URL}/api/csv/clear`, { method: 'POST' });
262 |             const data = await res.json();
263 |
264 |             if (res.ok) {
265 |                 try {
266 |                     localStorage.clear();
267 |                     sessionStorage.clear();
268 |                 } catch (e) {}
269 |
270 |                 closeClearModal();
271 |                 showToast("Database and records cleared successfully!");
272 |                 fetchStats();
273 |             } else {
274 |                 showToast(data.message || "Failed to clear database", true);
275 |             }
276 |         } catch (err) {
277 |             showToast("Network error: Could not reach server", true);
278 |         } finally {
279 |             btn.disabled = false;
280 |             btn.innerHTML = '<span class="material-icons text-sm mr-1.5">delete_forever</span> Yes, C
281 |
282 |     }
283 |
284 |     async function executeDeleteSession() {
285 |         const btn = document.getElementById('confirmDeleteSessionBtn');
286 |         btn.disabled = true;
287 |         btn.innerHTML = '<span class="material-icons text-sm mr-1.5 animate-spin">refresh</span> Dele
288 |
289 |         try {
290 |             const res = await fetch(`${BACKEND_URL}/api/whatsapp/delete_session`, { method: 'POST' });
291 |             const data = await res.json();
292 |
293 |             if (res.ok) {
294 |                 closeDeleteSessionModal();
295 |                 showToast("WhatsApp session deleted! Re-pair via QR code.");
296 |             } else {
297 |                 showToast(data.message || "Failed to delete session", true);
298 |             }
299 |         } catch (err) {
300 |             showToast("Network error: Could not reach server", true);
301 |         } finally {
302 |             btn.disabled = false;
303 |             btn.innerHTML = '<span class="material-icons text-sm mr-1.5">delete_forever</span> Yes, D
304 |         }
305 |     }
306 |
307 |     fetchStats();
308 |     setInterval(fetchStats, 10000);
309 | </script>
310 | </body>
311 |
312 | </html>

```

```

1 | <!DOCTYPE html>
2 | <html lang="en">
3 |
4 | <head>
5 |   <meta charset="UTF-8">
6 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7 |   <title>Live Dashboard</title>
8 |   <script src="https://cdn.tailwindcss.com"></script>
9 |   <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap" rel="st
10 |   <link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet">
11 |   <style>
12 |     body {
13 |       font-family: 'Inter', sans-serif;
14 |     }
15 |   </style>
16 |   <script src="config.js"></script>
17 |   <script src="auth.js"></script>
18 | </head>
19 |
20 | <body class="bg-slate-50 text-slate-900 min-h-screen p-6">
21 |   <div class="max-w-7xl mx-auto">
22 |     <!-- Header -->
23 |     <header class="flex flex-col sm:flex-row justify-between items-start sm:items-center gap-4 mb-8">
24 |       <div>
25 |         <h1 class="text-2xl font-bold text-slate-900">Live Sessions</h1>
26 |         <p class="text-slate-500 text-sm">Real-time participant tracking</p>
27 |       </div>
28 |       <div class="flex gap-3">
29 |         <a href="index.html"
30 |           class="bg-white border border-slate-300 text-slate-700 hover:bg-slate-50 px-4 py-2 ro
31 |           <span class="material-icons text-sm mr-2">arrow_back</span> New Pass
32 |         </a>
33 |         <button onclick="logout()"
34 |           class="bg-red-50 border border-red-200 text-red-600 hover:bg-red-100 px-4 py-2 rounde
35 |           <span class="material-icons text-sm mr-1.5">logout</span> Logout
36 |         </button>
37 |       </div>
38 |     </header>
39 |
40 |     <!-- Stats Overview -->
41 |
42 |     <div class="grid grid-cols-1 md:grid-cols-2 gap-6 mb-8">
43 |       <div class="bg-white p-6 rounded-xl border border-slate-200 shadow-sm flex items-center justi
44 |         <div>
45 |           <div class="text-slate-500 text-xs font-semibold uppercase tracking-wide">Active Part
46 |           <div class="text-3xl font-bold text-slate-900 mt-1" id="activeCount">0</div>
47 |         </div>
48 |         <div class="p-3 bg-blue-50 rounded-full text-blue-600">
49 |           <span class="material-icons">people</span>
50 |         </div>
51 |       </div>
52 |       <div class="bg-white p-6 rounded-xl border border-slate-200 shadow-sm flex items-center justi
53 |         <div>
54 |           <div class="text-slate-500 text-xs font-semibold uppercase tracking-wide">System Stat
55 |           <div class="mt-1 flex items-center gap-2">
56 |             <span class="w-3 h-3 bg-green-500 rounded-full animate-pulse"></span>
57 |             <span class="font-bold text-slate-700">Online</span>
58 |           </div>
59 |         </div>
60 |         <div class="p-3 bg-green-50 rounded-full text-green-600">
61 |           <span class="material-icons">dns</span>
62 |         </div>
63 |       </div>
64 |     </div>
65 |
66 |     <!-- Participants Table -->
67 |     <div class="bg-white rounded-xl shadow-sm border border-slate-200 overflow-hidden">
68 |       <div class="overflow-x-auto">
69 |         <table class="w-full text-left">
70 |           <thead class="bg-slate-50 border-b border-slate-200">
71 |             <tr>
72 |               <th class="px-6 py-4 text-xs font-semibold text-slate-500 uppercase tracking-
73 |               </th>
74 |               <th class="px-6 py-4 text-xs font-semibold text-slate-500 uppercase tracking-
75 |               </th>
76 |               <th class="px-6 py-4 text-xs font-semibold text-slate-500 uppercase tracking-
77 |               </th>
78 |               <th class="px-6 py-4 text-xs font-semibold text-slate-500 uppercase tracking-
79 |               Time</th>
80 |               <th class="px-6 py-4 text-xs font-semibold text-slate-500 uppercase tracking-

```



```
81 |                 Remaining</th>
82 |                 <th class="px-6 py-4 text-xs font-semibold text-slate-500 uppercase tracking-
83 |                 </th>
84 |             </tr>
85 |         </thead>
86 |         <tbody id="sessionsBody" class="divide-y divide-slate-100">
87 |             <!-- Rows injected via JS -->
88 |         </tbody>
89 |     </table>
90 | </div>
91 |
92 |     <!-- Empty State -->
93 |     <div id="emptyState" class="hidden p-12 text-center text-slate-500">
94 |         <span class="material-icons text-4xl text-slate-300 mb-2">event_busy</span>
95 |         <p>No active sessions currently.</p>
96 |     </div>
97 | </div>
98 | </div>
99 |
100 | <script>
101 |     async function fetchSessions() {
102 |         try {
103 |             const response = await fetch(`${BACKEND_URL}/api/sessions`);
104 |             if (!response.ok) return;
105 |             const sessions = await response.json();
106 |             renderTable(sessions);
107 |         } catch (e) {
108 |             console.error("Failed to fetch sessions", e);
109 |         }
110 |     }
111 |
112 |     function formatTime(seconds) {
113 |         if (seconds <= 0) return "00:00";
114 |         const m = Math.floor(seconds / 60).toString().padStart(2, '0');
115 |         const s = (seconds % 60).toString().padStart(2, '0');
116 |         return `${m}:${s}`;
117 |     }
118 |
119 |     function renderTable(sessions) {
120 |         const tbody = document.getElementById('sessionsBody');
```

```

121 |     const emptyState = document.getElementById('emptyState');
122 |     const activeCount = document.getElementById('activeCount');
123 |
124 |     tbody.innerHTML = '';
125 |
126 |     // Count actual active sessions
127 |     const active = sessions.filter(s => s.remaining_seconds > 0).length;
128 |     activeCount.innerText = active;
129 |
130 |     if (sessions.length === 0) {
131 |         emptyState.classList.remove('hidden');
132 |     } else {
133 |         emptyState.classList.add('hidden');
134 |
135 |         sessions.forEach(session => {
136 |             const row = document.createElement('tr');
137 |             row.className = "hover:bg-slate-50 transition-colors";
138 |
139 |             // Determine Status Style
140 |             let statusClass = "bg-slate-100 text-slate-600";
141 |             let timerColor = "text-blue-600";
142 |
143 |             if (session.status === "Warning") {
144 |                 statusClass = "bg-yellow-100 text-yellow-800";
145 |                 timerColor = "text-yellow-600";
146 |             }
147 |             if (session.status === "Ended") {
148 |                 statusClass = "bg-red-100 text-red-800";
149 |                 timerColor = "text-red-600";
150 |             }
151 |             if (session.status === "Active") {
152 |                 statusClass = "bg-green-100 text-green-800";
153 |                 timerColor = "text-blue-600";
154 |             }
155 |
156 |             row.innerHTML = `
157 |                 <td class="px-6 py-4 font-medium text-slate-900">${session.name || '-'}</td>
158 |                 <td class="px-6 py-4 text-slate-500 font-mono text-xs">${session.phone}</td>
159 |                 <td class="px-6 py-4 text-slate-500">${session.duration}m</td>
160 |                 <td class="px-6 py-4 text-slate-500 font-mono text-xs">${session.start_time}</td>
161 |                 <td class="px-6 py-4 font-mono font-bold ${timerColor}">
162 |                     ${formatTime(session.remaining_seconds)}
163 |                 </td>
164 |                 <td class="px-6 py-4">
165 |                     <span class="inline-flex items-center px-2.5 py-0.5 rounded-full text-xs font
166 |                         ${session.status}
167 |                     </span>
168 |                 </td>
169 |             `;
170 |             tbody.appendChild(row);
171 |         });
172 |     }
173 | }
174 |
175 | // Initial Call
176 | fetchSessions();
177 |
178 | // Refresh every 1 second
179 | setInterval(fetchSessions, 1000);
180 | </script>
181 | </body>
182 |
183 | </html>

```

File: scan_result.html — QR Verification & Pass Validation Gate Screen

```

1 | <!DOCTYPE html>
2 | <html lang="en">
3 |
4 | <head>
5 |   <meta charset="UTF-8">
6 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7 |   <title>Verification Result</title>
8 |   <script src="https://cdn.tailwindcss.com"></script>
9 |   <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap" rel="st
10 |   <link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet">
11 |   <style>
12 |     body {
13 |       font-family: 'Inter', sans-serif;
14 |     }
15 |
16 |     @keyframes scaleIn {
17 |       0% {
18 |         transform: scale(0);
19 |         opacity: 0;
20 |       }
21 |
22 |       80% {
23 |         transform: scale(1.1);
24 |         opacity: 1;
25 |       }
26 |
27 |       100% {
28 |         transform: scale(1);
29 |         opacity: 1;
30 |       }
31 |     }
32 |
33 |     .animate-pop {
34 |       animation: scaleIn 0.5s ease-out forwards;
35 |     }
36 |   </style>
37 |   <script src="config.js"></script>
38 | </head>
39 |
40 | <body class="bg-slate-50 min-h-screen flex items-center justify-center p-4">

```

```

41 |
42 |   <div class="max-w-md w-full bg-white rounded-xl shadow-lg overflow-hidden border border-slate-200">
43 |
44 |     {% if status == "success" %}
45 |     <!-- Success Content -->
46 |     <div class="p-8 text-center">
47 |       <div class="w-20 h-20 bg-green-100 rounded-full flex items-center justify-center mx-auto mb-6">
48 |         <span class="material-icons text-5xl text-green-600">check_circle</span>
49 |       </div>
50 |
51 |       <h1 class="text-2xl font-bold text-slate-900 mb-2">Verified Successfully</h1>
52 |       <p class="text-slate-500 text-sm mb-6">Welcome, your session details are below.</p>
53 |
54 |       <div class="bg-slate-50 rounded-lg p-4 text-left border border-slate-200 space-y-3">
55 |         <div class="flex justify-between">
56 |           <span class="text-xs font-medium text-slate-500 uppercase">Participant</span>
57 |           <span class="text-sm font-semibold text-slate-900">{{ name }}</span>
58 |         </div>
59 |         <div class="flex justify-between">
60 |           <span class="text-xs font-medium text-slate-500 uppercase">Phone</span>
61 |           <span class="text-sm font-mono text-slate-700">{{ phone }}</span>
62 |         </div>
63 |         <div class="flex justify-between">
64 |           <span class="text-xs font-medium text-slate-500 uppercase">Pass Type</span>
65 |           <span class="text-sm font-semibold text-slate-900">{{ plan }}</span>
66 |         </div>
67 |         <div class="flex justify-between">
68 |           <span class="text-xs font-medium text-slate-500 uppercase">Duration</span>
69 |           <span class="text-sm font-semibold text-blue-600">{{ duration }} Mins</span>
70 |         </div>
71 |         <div class="pt-3 border-t border-slate-200 flex justify-between">
72 |           <span class="text-xs font-medium text-slate-500 uppercase">Entry ID</span>
73 |           <span class="text-xs font-mono text-slate-600">{{ transaction_id }}</span>
74 |         </div>
75 |       </div>
76 |
77 |       <div class="mt-8">
78 |         <!-- Start Timer Form -->
79 |         <form id="startTimerForm" class="mt-4">
80 |           <input type="hidden" name="name" value="{{ name }}">

```

```
81 | <input type="hidden" name="phone" value="{{ phone }}">
82 | <input type="hidden" name="transaction_id" value="{{ transaction_id }}">
83 | <input type="hidden" name="duration" value="{{ duration }}">
84 |
85 | <button type="submit" id="startTimerBtn"
86 |     class="w-full bg-blue-600 hover:bg-blue-700 text-white font-medium py-3 rounded-l
87 |     <span class="material-icons text-sm mr-2">timer</span> Start {{ duration }} Min T
88 | </button>
89 | </form>
90 | <div id="timerMessage" class="hidden mt-3 text-sm font-medium"></div>
91 | </div>
92 |
93 | <script>
94 |     document.getElementById('startTimerForm').addEventListener('submit', async (e) => {
95 |         e.preventDefault();
96 |         const btn = document.getElementById('startTimerBtn');
97 |         const msg = document.getElementById('timerMessage');
98 |         const formData = new FormData(e.target);
99 |         const tid = formData.get('transaction_id');
100 |         const durationStr = formData.get('duration');
101 |
102 |         // Optimistic UI Update
103 |         btn.disabled = true;
104 |         btn.innerHTML = '<span class="material-icons text-sm mr-2 animate-spin">refresh</span
105 |
106 |         try {
107 |             // Attempt to call server
108 |             let response;
109 |             let result;
110 |             let serverSuccess = false;
111 |
112 |             try {
113 |                 // Only try fetch if online using AbortController for quick timeout?
114 |                 // Or just let it fail.
115 |                 response = await fetch(`${BACKEND_URL}/start_timer`, {
116 |                     method: 'POST',
117 |                     body: formData
118 |                 });
119 |                 if (response.ok) {
120 |                     result = await response.json();
```

```

121 |         serverSuccess = true;
122 |     } else {
123 |         result = await response.json();
124 |         throw new Error(result.message || 'Server Error');
125 |     }
126 | } catch (err) {
127 |     console.log("Server unreachable or error:", err);
128 |     // Fallback logic could go here if we want to allow starting timer OFFLINE?
129 |     // But usually we need server to validate access.
130 |     // User asked for "timer continues to work... if server crashes".
131 |     // This implies the timer ALREADY started.
132 |     // If user clicks start while offline, maybe we should block or allow?
133 |     // Let's assume we BLOCK starting new sessions if offline,
134 |     // but we RESTORE existing ones.
135 |
136 |     // Re-throwing to show error for now, as new session requires server validation
137 |     throw err;
138 | }
139 |
140 | msg.classList.remove('hidden');
141 | if (serverSuccess) {
142 |     msg.className = "mt-3 text-sm font-medium text-green-600";
143 |     msg.textContent = "Timer Started Successfully!";
144 |     btn.innerHTML = '<span class="material-icons text-sm mr-2">check</span> Start';
145 |
146 |     // SAVE KEY TO LOCAL STORAGE
147 |     if (result.restore_key) {
148 |         localStorage.setItem(`restore_key_${tid}`, result.restore_key);
149 |     }
150 |
151 |     // Start Client Side Countdown
152 |     // PREFER Server provided end time, but fallback to local calcs
153 |     let endTimeIso = result.end_time;
154 |
155 |     // Save End Time to LocalStorage for Offline Persistence
156 |     localStorage.setItem(`session_end_time_${tid}`, endTimeIso);
157 |     localStorage.setItem(`session_duration_${tid}`, result.duration);
158 |
159 |     startCountdown(endTimeIso);
160 | }

```

```

161 |
162 |     } catch (error) {
163 |         msg.classList.remove('hidden');
164 |         msg.className = "mt-3 text-sm font-medium text-red-600";
165 |         msg.textContent = "Error: " + error.message;
166 |         btn.disabled = false;
167 |         btn.innerHTML = '<span class="material-icons text-sm mr-2">timer</span> Retry Sta
168 |     }
169 | });
170 |
171 | // Auto-Restore Timer from LocalStorage on Load
172 | // This runs even if server is down!
173 | (function () {
174 |     const tid = "{{ transaction_id }}"; // Jinja2 Variable
175 |
176 |     const storedEndTime = localStorage.getItem(`session_end_time_${tid}`);
177 |
178 |     if (storedEndTime) {
179 |         // We have an active local session!
180 |         console.log("Restoring session from LocalStorage");
181 |         const form = document.getElementById('startTimerForm');
182 |         const btn = document.getElementById('startTimerBtn');
183 |
184 |         // Hide button / Show countdown immediately
185 |         startCountdown(storedEndTime);
186 |     }
187 | })();
188 |
189 | function startCountdown(endTimeIso) {
190 |     const btn = document.getElementById('startTimerBtn');
191 |     const form = document.getElementById('startTimerForm');
192 |
193 |     // If form exists (it might not if we already replaced it), modify it
194 |     if (form) {
195 |         form.innerHTML = `
196 |             <div class="mt-6 text-center">
197 |                 <div class="text-xs text-slate-500 uppercase font-semibold mb-1">Time Rem
198 |                 <div id="countdownDisplay" class="text-4xl font-mono font-bold text-blue-
199 |                     --:--
200 |                 </div>

```

```
201 |         <div class="text-xs text-slate-400 mt-2">Session ends at ${new Date(endTi
202 |             </div>
203 |         `;
204 |     }
205 |
206 |     const endTime = new Date(endTimeIso).getTime();
207 |
208 |     // Update function
209 |     const updateTimer = () => {
210 |         const now = new Date().getTime();
211 |         const distance = endTime - now;
212 |
213 |         const display = document.getElementById('countdownDisplay');
214 |         if (!display) return; // Element removed?
215 |
216 |         if (distance < 0) {
217 |             clearInterval(window.timerInterval); // Use global/window var to track
218 |             display.innerHTML = "SESSION ENDED";
219 |             display.classList.replace('text-blue-600', 'text-red-600');
220 |
221 |             // Cleanup LocalStorage? Maybe keep for history or clear on explicit exit.
222 |             // localStorage.removeItem(`session_end_time_${tid}`);
223 |             return;
224 |         }
225 |
226 |         const minutes = Math.floor((distance % (1000 * 60 * 60)) / (1000 * 60));
227 |         const seconds = Math.floor((distance % (1000 * 60)) / 1000);
228 |
229 |         display.innerHTML =
230 |             (minutes < 10 ? "0" + minutes : minutes) + ":" +
231 |             (seconds < 10 ? "0" + seconds : seconds);
232 |     };
233 |
234 |     // Start Interval
235 |     updateTimer(); // Run once immediately
236 |     window.timerInterval = setInterval(updateTimer, 1000);
237 | }
238 | </script>
239 | </div>
240 |
```

```

241 |     {% elif status == "check_restore" %}
242 |     <!-- Restore Check State (Initially Hidden or Loading) -->
243 |     <div class="p-8 text-center" id="restoreContainer">
244 |         <div class="w-16 h-16 border-4 border-blue-200 border-t-blue-600 rounded-full animate-spin mx
245 |         </div>
246 |         <h2 class="text-lg font-semibold text-slate-700">Checking Session...</h2>
247 |         <p class="text-sm text-slate-500">Verifying your browser key...</p>
248 |     </div>
249 |
250 |     <!-- Hidden Inputs for Restore Logic -->
251 |     <input type="hidden" id="restoreTid" value="{ transaction_id }">
252 |
253 |     <script>
254 |         // Auto-run on load
255 |         (async function () {
256 |             const tid = document.getElementById('restoreTid').value;
257 |             const key = localStorage.getItem(`restore_key_${tid}`);
258 |             const container = document.getElementById('restoreContainer');
259 |
260 |             if (!key) {
261 |                 container.innerHTML = `
262 |                     <div class="w-20 h-20 bg-yellow-100 rounded-full flex items-center justify-center
263 |                     <span class="material-icons text-5xl text-yellow-600">lock_clock</span>
264 |                     </div>
265 |                     <h1 class="text-2xl font-bold text-slate-900 mb-2">Session Active</h1>
266 |                     <p class="text-slate-500 text-sm mb-6">This session is currently running on anoth
267 |                     <div class="bg-yellow-50 text-yellow-700 p-4 rounded-lg text-xs border border-ye
268 |                         If this is you, please continue on the original tab.
269 |                     </div>
270 |                 `;
271 |                 return;
272 |             }
273 |
274 |             try {
275 |                 const response = await fetch(`${BACKEND_URL}/api/verify_restore`, {
276 |                     method: 'POST',
277 |                     headers: { 'Content-Type': 'application/json' },
278 |                     body: JSON.stringify({ transaction_id: tid, restore_key: key })
279 |                 });
280 |

```

```

281 |                 const result = await response.json();
282 |
283 |                 if (response.ok && result.status === 'restored') {
284 |                     // Success! Restore the Timer UI
285 |                     const durationMinutes = result.duration;
286 |                     const endTimeIso = result.end_time;
287 |
288 |                     // Inject the timer UI directly
289 |                     container.innerHTML = `
290 |                         <div class="w-20 h-20 bg-green-100 rounded-full flex items-center justify-ce
291 |                         <span class="material-icons text-5xl text-green-600">check_circle</span>
292 |                         </div>
293 |                         <h1 class="text-2xl font-bold text-slate-900 mb-2">Session Restored</h1>
294 |
295 |                         <div class="mt-6 text-center">
296 |                             <div class="text-xs text-slate-500 uppercase font-semibold mb-1">Time Rem
297 |                             <div id="countdownDisplay" class="text-4xl font-mono font-bold text-blue-
298 |                                 ---
299 |                             </div>
300 |                             <div class="text-xs text-slate-400 mt-2">Session ends at ${new Date(endT
301 |                         </div>
302 |                     `;
303 |
304 |                     startCountdown(endTimeIso);
305 |
306 |                 } else {
307 |                     container.innerHTML = `
308 |                         <div class="w-20 h-20 bg-red-100 rounded-full flex items-center justify-cent
309 |                         <span class="material-icons text-5xl text-red-600">running_with_errors</s
310 |                         </div>
311 |                         <h1 class="text-2xl font-bold text-slate-900 mb-2">Restore Failed</h1>
312 |                         <p class="text-slate-500 text-sm mb-6">${result.message || 'Session expired o
313 |                     `;
314 |                 }
315 |
316 |             } catch (e) {
317 |                 container.innerHTML = `
318 |                     <h1 class="text-xl font-bold text-red-600">Network Error</h1>
319 |                     <p class="text-sm tex-slate-500">Could not verify session.</p>
320 |                 `;

```

```
321 |     }
322 |   })();
323 |
324 |   function startCountdown(endTimeIso) {
325 |     const endTime = new Date(endTimeIso).getTime();
326 |
327 |     const timerInterval = setInterval(() => {
328 |       const now = new Date().getTime();
329 |       const distance = endTime - now;
330 |
331 |       if (distance < 0) {
332 |         clearInterval(timerInterval);
333 |         const display = document.getElementById('countdownDisplay');
334 |         if (display) {
335 |           display.innerHTML = "SESSION ENDED";
336 |           display.classList.replace('text-blue-600', 'text-red-600');
337 |         }
338 |         return;
339 |       }
340 |
341 |       const minutes = Math.floor((distance % (1000 * 60 * 60)) / (1000 * 60));
342 |       const seconds = Math.floor((distance % (1000 * 60)) / 1000);
343 |
344 |       const display = document.getElementById('countdownDisplay');
345 |       if (display) {
346 |         display.innerHTML =
347 |           (minutes < 10 ? "0" + minutes : minutes) + ":" +
348 |           (seconds < 10 ? "0" + seconds : seconds);
349 |       }
350 |
351 |     }, 1000);
352 |   }
353 | </script>
354 |
355 | {% else %}
356 | <!-- Error Content -->
357 | <div class="p-8 text-center">
358 |   <div class="w-20 h-20 bg-red-100 rounded-full flex items-center justify-center mx-auto mb-6">
359 |     <span class="material-icons text-5xl text-red-600">error</span>
360 |   </div>
```



```
361 |
362 |         <h1 class="text-2xl font-bold text-slate-900 mb-2">Verification Failed</h1>
363 |         <p class="text-slate-500 text-sm mb-6">{{ message }}</p>
364 |
365 |         <div class="bg-red-50 text-red-700 p-4 rounded-lg text-xs border border-red-100">
366 |             Please try scanning the QR code again or contact the administrator.
367 |         </div>
368 |     </div>
369 |     {% endif %}
370 |
371 | </div>
372 |
373 | <div id="offlineIndicator"
374 |     class="fixed bottom-4 right-4 bg-slate-800 text-white px-4 py-2 rounded-lg text-xs font-medium hi
375 |     You are offline. Timer is running locally.
376 | </div>
377 |
378 | <script>
379 |     // Register Service Worker
380 |     if ('serviceWorker' in navigator) {
381 |         window.addEventListener('load', () => {
382 |             navigator.serviceWorker.register('/static/sw.js')
383 |                 .then(registration => {
384 |                     console.log('ServiceWorker registration successful with scope: ', registration.sc
385 |                 })
386 |                 .catch(err => {
387 |                     console.log('ServiceWorker registration failed: ', err);
388 |                 });
389 |         });
390 |     }
391 |
392 |     // Offline Indicator
393 |     window.addEventListener('online', updateOnlineStatus);
394 |     window.addEventListener('offline', updateOnlineStatus);
395 |
396 |     function updateOnlineStatus() {
397 |         const indicator = document.getElementById('offlineIndicator');
398 |         if (!navigator.onLine) {
399 |             indicator.classList.remove('hidden');
400 |         } else {
401 |             indicator.classList.add('hidden');
402 |         }
403 |     }
404 |     updateOnlineStatus(); // Initial check
405 | </script>
406 | </body>
407 |
408 | </html>
```

File: health.html — System Health Monitoring & Server Diagnostics Interface

```

1 | <!DOCTYPE html>
2 | <html lang="en">
3 |
4 | <head>
5 |   <meta charset="UTF-8">
6 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7 |   <title>System Health</title>
8 |   <script src="https://cdn.tailwindcss.com"></script>
9 |   <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap" rel="st
10 |   <link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet">
11 |   <style>
12 |     body {
13 |       font-family: 'Inter', sans-serif;
14 |     }
15 |   </style>
16 |   <script src="config.js"></script>
17 | </head>
18 |
19 | <body class="bg-slate-100 min-h-screen flex items-center justify-center p-4">
20 |
21 |   {% if not authenticated %}
22 |   <!-- Login Form -->
23 |   <div class="max-w-sm w-full bg-white rounded-xl shadow-lg p-8 border border-slate-200">
24 |     <h2 class="text-2xl font-bold text-slate-800 mb-6 text-center">Admin Login</h2>
25 |
26 |     {% if error %}
27 |     <div class="mb-4 p-3 bg-red-50 text-red-700 text-sm rounded-lg border border-red-100">
28 |       {{ error }}
29 |     </div>
30 |     {% endif %}
31 |
32 |     <form action="/health/login" method="POST" class="space-y-4">
33 |       <div>
34 |         <label class="block text-sm font-medium text-slate-700 mb-1">Login ID</label>
35 |         <input type="text" name="username" required
36 |           class="w-full px-4 py-2 border border-slate-300 rounded-lg focus:ring-2 focus:ring-bl
37 |       </div>
38 |       <div>
39 |         <label class="block text-sm font-medium text-slate-700 mb-1">Password</label>
40 |         <input type="password" name="password" required
41 |           class="w-full px-4 py-2 border border-slate-300 rounded-lg focus:ring-2 focus:ring-bl
42 |       </div>
43 |       <button type="submit"
44 |         class="w-full bg-blue-600 hover:bg-blue-700 text-white font-semibold py-2 rounded-lg tran
45 |         Access Dashboard
46 |       </button>
47 |     </form>
48 |   </div>
49 |
50 |   {% else %}
51 |   <!-- Dashboard -->
52 |   <div class="max-w-4xl w-full">
53 |
54 |     <header class="flex justify-between items-center mb-8">
55 |       <div>
56 |         <h1 class="text-3xl font-bold text-slate-800">System Health</h1>
57 |         <p class="text-slate-500">Real-time server monitoring</p>
58 |       </div>
59 |       <a href="/health/logout" class="text-sm text-red-600 hover:text-red-700 font-medium">Logout</
60 |     </header>
61 |
62 |     <div class="grid grid-cols-1 md:grid-cols-2 gap-6">
63 |
64 |       <!-- Active Sessions Card -->
65 |       <div class="bg-white rounded-xl shadow p-6 border border-slate-200 flex items-center">
66 |         <div class="w-12 h-12 rounded-full bg-blue-100 flex items-center justify-center mr-4">
67 |           <span class="material-icons text-blue-600">devices</span>
68 |         </div>
69 |         <div>
70 |           <h3 class="text-sm font-medium text-slate-500 uppercase">Active Sessions</h3>
71 |           <p class="text-3xl font-bold text-slate-900" id="activeCount">--</p>
72 |         </div>
73 |       </div>
74 |
75 |       <!-- Connection Strength Card -->
76 |       <div class="bg-white rounded-xl shadow p-6 border border-slate-200 flex items-center">
77 |         <div class="w-12 h-12 rounded-full bg-green-100 flex items-center justify-center mr-4" id
78 |           <span class="material-icons text-green-600" id="pingIcon">wifi</span>
79 |         </div>
80 |       </div>

```

```

81 |         <h3 class="text-sm font-medium text-slate-500 uppercase">Connection Strength</h3>
82 |         <div class="flex items-baseline space-x-2">
83 |             <p class="text-3xl font-bold text-slate-900" id="pingStatus">Checking...</p>
84 |             <span class="text-sm text-slate-400 font-mono" id="pingMs">-- ms</span>
85 |         </div>
86 |     </div>
87 | </div>
88 |
89 | </div>
90 |
91 | <!-- Service Status Table -->
92 | <div class="mt-6 bg-white rounded-xl shadow border border-slate-200 overflow-hidden">
93 |     <div class="px-6 py-4 border-b border-slate-200">
94 |         <h3 class="text-lg font-semibold text-slate-800">Service Status</h3>
95 |     </div>
96 |     <table class="w-full text-left">
97 |         <thead class="bg-slate-50 border-b border-slate-200">
98 |             <tr>
99 |                 <th class="px-6 py-3 text-xs font-semibold text-slate-500 uppercase">Endpoint</th>
100 |                 <th class="px-6 py-3 text-xs font-semibold text-slate-500 uppercase">Status</th>
101 |                 <th class="px-6 py-3 text-xs font-semibold text-slate-500 uppercase">Latency</th>
102 |             </tr>
103 |         </thead>
104 |         <tbody class="divide-y divide-slate-100">
105 |             <!-- Home -->
106 |             <tr>
107 |                 <td class="px-6 py-4 font-medium text-slate-700">Home (/)</td>
108 |                 <td class="px-6 py-4" id="status-home">Checking...</td>
109 |                 <td class="px-6 py-4 font-mono text-xs text-slate-500" id="latency-home">--</td>
110 |             </tr>
111 |             <!-- Participants -->
112 |             <tr>
113 |                 <td class="px-6 py-4 font-medium text-slate-700">Participants (/participants)</td>
114 |                 <td class="px-6 py-4" id="status-parts">Checking...</td>
115 |                 <td class="px-6 py-4 font-mono text-xs text-slate-500" id="latency-parts">--</td>
116 |             </tr>
117 |             <!-- Scan Data -->
118 |             <tr>
119 |                 <td class="px-6 py-4 font-medium text-slate-700">Active Sessions Data (/api/sessi
120 |                 <td class="px-6 py-4" id="status-api">Checking...</td>

```

```

121 |         <td class="px-6 py-4 font-mono text-xs text-slate-500" id="latency-api">--</td>
122 |     </tr>
123 | </tbody>
124 | </table>
125 | </div>
126 |
127 | <!-- Connection Log / Details -->
128 | <div class="mt-6 bg-white rounded-xl shadow p-6 border border-slate-200">
129 |   <h3 class="text-lg font-semibold text-slate-800 mb-4">Server Info</h3>
130 |   <div class="flex items-center space-x-2 text-sm text-slate-600">
131 |     <span class="w-2 h-2 rounded-full bg-green-500 animate-pulse"></span>
132 |     <span>Server is running and listening for connections.</span>
133 |   </div>
134 |   <p class="mt-2 text-xs text-slate-400">Server Time: <span id="serverTime">--</span></p>
135 | </div>
136 |
137 | </div>
138 |
139 | <script>
140 |   async function checkEndpoint(url, statusId, latencyId) {
141 |     const start = performance.now();
142 |     try {
143 |       const response = await fetch(url, { method: 'GET' }); // Changed to GET to avoid 405
144 |       const end = performance.now();
145 |       const latency = Math.round(end - start);
146 |
147 |       document.getElementById(latencyId).innerText = latency + " ms";
148 |       const statusEl = document.getElementById(statusId);
149 |
150 |       if (response.ok) {
151 |         statusEl.innerHTML = '<span class="inline-flex items-center px-2.5 py-0.5 rounded-ful
152 |       } else {
153 |         statusEl.innerHTML = '<span class="inline-flex items-center px-2.5 py-0.5 rounded-ful
154 |       }
155 |       return latency; // Return for overall health calculation if needed
156 |     } catch (e) {
157 |       document.getElementById(latencyId).innerText = "--";
158 |       document.getElementById(statusId).innerHTML = '<span class="inline-flex items-center px-2
159 |       return 9999;
160 |     }
161 |   }
162 |
163 |   async function updateStats() {
164 |     // 1. Get Overall Stats
165 |     const start = performance.now();
166 |     try {
167 |       const response = await fetch(`${BACKEND_URL}/api/health_stats`);
168 |       const end = performance.now();
169 |       const latency = Math.round(end - start);
170 |
171 |       if (response.ok) {
172 |         const data = await response.json();
173 |
174 |         // Update Active Count
175 |         document.getElementById('activeCount').innerText = data.active_sessions;
176 |         document.getElementById('serverTime').innerText = new Date(data.server_time).toLocale
177 |
178 |         // Update Latency UI
179 |         const pingMs = document.getElementById('pingMs');
180 |         const pingStatus = document.getElementById('pingStatus');
181 |         const icon = document.getElementById('pingIcon');
182 |         const iconBg = document.getElementById('pingIconBg');
183 |
184 |         pingMs.innerText = latency + " ms";
185 |
186 |         if (latency < 100) {
187 |           pingStatus.innerText = "Excellent";
188 |           pingStatus.className = "text-3xl font-bold text-green-600";
189 |           icon.innerText = "wifi";
190 |           icon.className = "material-icons text-green-600";
191 |           iconBg.className = "w-12 h-12 rounded-full bg-green-100 flex items-center justify
192 |         } else if (latency < 300) {
193 |           pingStatus.innerText = "Good";
194 |           pingStatus.className = "text-3xl font-bold text-blue-600";
195 |           icon.innerText = "network_wifi";
196 |           icon.className = "material-icons text-blue-600";
197 |           iconBg.className = "w-12 h-12 rounded-full bg-blue-100 flex items-center justify-
198 |         } else {
199 |           pingStatus.innerText = "Weak";
200 |           pingStatus.className = "text-3xl font-bold text-yellow-600";

```

```

201 |             icon.innerText = "signal_wifi_bad";
202 |             icon.className = "material-icons text-yellow-600";
203 |             iconBg.className = "w-12 h-12 rounded-full bg-yellow-100 flex items-center justif
204 |         }
205 |     }
206 | } catch (e) {
207 |     document.getElementById('pingStatus').innerText = "Offline";
208 |     document.getElementById('pingStatus').className = "text-3xl font-bold text-red-600";
209 | }
210 |
211 | // 2. Ping Endpoints
212 | checkEndpoint('/', 'status-home', 'latency-home');
213 | checkEndpoint('/participants', 'status-parts', 'latency-parts');
214 | checkEndpoint('/api/sessions', 'status-api', 'latency-api');
215 | }
216 |
217 | // Initial Load
218 | updateStats();
219 |
220 | // Refresh every 3 seconds
221 | setInterval(updateStats, 3000);
222 | </script>
223 | {% endif %}
224 |
225 | </body>
226 |
227 | </html>

```

File: auth.js — Role-Based Route Guard & Local Storage Session Manager

```

1 | // auth.js - Admin Authentication Guard
2 |
3 | function getAuthToken() {
4 |     return localStorage.getItem("admin_token");
5 | }
6 |
7 | function isAuthenticated() {
8 |     const token = getAuthToken();
9 |     return token !== null && token !== "";
10 | }
11 |
12 | function requireAuth() {
13 |     const currentPath = window.location.pathname;
14 |     const isLoginPage = currentPath.endsWith("login.html") || currentPath.endsWith("login");
15 |
16 |     if (!isAuthenticated() && !isLoginPage) {
17 |         window.location.href = "login.html";
18 |     } else if (isAuthenticated() && isLoginPage) {
19 |         window.location.href = "index.html";
20 |     }
21 | }
22 |
23 | function logout() {
24 |     localStorage.removeItem("admin_token");
25 |     localStorage.removeItem("admin_user");
26 |     sessionStorage.clear();
27 |     window.location.href = "login.html";
28 | }
29 |
30 | // Run auth check immediately upon loading
31 | requireAuth();

```

File: config.js — Centralized API Endpoint Configuration

```

1 | const BACKEND_URL = "https://wt-project-q8sm.onrender.com";

```

File: static/style.css — Design System, Glassmorphism Tokens & Responsive Layouts

```
1 | :root {
2 |   --primary: #6366f1;
3 |   --primary-hover: #4f46e5;
4 |   --bg-dark: #0f172a;
5 |   --glass-bg: rgba(30, 41, 59, 0.7);
6 |   --glass-border: rgba(255, 255, 255, 0.1);
7 |   --text-main: #f8fafc;
8 |   --text-muted: #94a3b8;
9 | }
10 |
11 | * {
12 |   box-sizing: border-box;
13 |   margin: 0;
14 |   padding: 0;
15 | }
16 |
17 | body {
18 |   font-family: 'Inter', sans-serif;
19 |   background-color: var(--bg-dark);
20 |   background-image:
21 |     radial-gradient(at 0% 0%, hsla(253,16%,7%,1) 0, transparent 50%),
22 |     radial-gradient(at 50% 0%, hsla(225,39%,30%,1) 0, transparent 50%),
23 |     radial-gradient(at 100% 0%, hsla(339,49%,30%,1) 0, transparent 50%);
24 |   background-size: cover;
25 |   background-attachment: fixed;
26 |   min-height: 100vh;
27 |   display: flex;
28 |   justify-content: center;
29 |   align-items: center;
30 |   color: var(--text-main);
31 | }
32 |
33 | .glass-container {
34 |   background: var(--glass-bg);
35 |   backdrop-filter: blur(12px);
36 |   -webkit-backdrop-filter: blur(12px);
37 |   border: 1px solid var(--glass-border);
38 |   border-radius: 16px;
39 |   padding: 3rem;
40 |   width: 100%;
41 |   max-width: 500px;
42 |   box-shadow: 0 4px 30px rgba(0, 0, 0, 0.5);
43 | }
44 |
45 | header {
46 |   text-align: center;
47 |   margin-bottom: 2.5rem;
48 | }
49 |
50 | h1 {
51 |   font-size: 2.5rem;
52 |   font-weight: 700;
53 |   margin-bottom: 0.5rem;
54 |   letter-spacing: -0.05em;
55 | }
56 |
57 | .highlight {
58 |   color: var(--primary);
59 |   background: linear-gradient(to right, #818cf8, #6366f1);
60 |   -webkit-background-clip: text;
61 |   -webkit-text-fill-color: transparent;
62 | }
63 |
64 | p {
65 |   color: var(--text-muted);
66 |   font-size: 0.95rem;
67 | }
68 |
69 | .input-group {
70 |   margin-bottom: 1.5rem;
71 |   display: flex;
72 |   flex-direction: column;
73 | }
74 |
75 | .row {
76 |   display: flex;
77 |   gap: 1rem;
78 | }
79 |
80 | .row .input-group {
```

```
81 |     flex: 1;
82 | }
83 |
84 | label {
85 |     font-size: 0.85rem;
86 |     font-weight: 500;
87 |     margin-bottom: 0.5rem;
88 |     color: var(--text-muted);
89 | }
90 |
91 | input {
92 |     background: rgba(15, 23, 42, 0.6);
93 |     border: 1px solid var(--glass-border);
94 |     border-radius: 8px;
95 |     padding: 0.75rem 1rem;
96 |     color: white;
97 |     font-size: 1rem;
98 |     transition: all 0.2s;
99 |     font-family: inherit;
100 | }
101 |
102 | input:focus {
103 |     outline: none;
104 |     border-color: var(--primary);
105 |     box-shadow: 0 0 0 2px rgba(99, 102, 241, 0.2);
106 | }
107 |
108 | button {
109 |     width: 100%;
110 |     padding: 1rem;
111 |     border: none;
112 |     border-radius: 8px;
113 |     background: var(--primary);
114 |     color: white;
115 |     font-size: 1rem;
116 |     font-weight: 600;
117 |     cursor: pointer;
118 |     transition: all 0.2s;
119 |     display: flex;
120 |     justify-content: center;
```

```
121 |     align-items: center;
122 |     gap: 0.5rem;
123 | }
124 |
125 | button:hover {
126 |     background: var(--primary-hover);
127 |     transform: translateY(-1px);
128 | }
129 |
130 | button:disabled {
131 |     opacity: 0.7;
132 |     cursor: not-allowed;
133 |     transform: none;
134 | }
135 |
136 | .loader {
137 |     width: 16px;
138 |     height: 16px;
139 |     border: 2px solid #ffffff;
140 |     border-bottom-color: transparent;
141 |     border-radius: 50%;
142 |     display: none;
143 |     animation: rotation 1s linear infinite;
144 | }
145 |
146 | .loading .loader {
147 |     display: inline-block;
148 | }
149 |
150 | @keyframes rotation {
151 |     0% { transform: rotate(0deg); }
152 |     100% { transform: rotate(360deg); }
153 | }
154 |
155 | .status-area {
156 |     margin-top: 1.5rem;
157 |     text-align: center;
158 |     font-size: 0.9rem;
159 |     min-height: 1.2em;
160 | }
161 |
162 | /* Mobile responsive */
163 | @media (max-width: 480px) {
164 |     .glass-container {
165 |         padding: 2rem;
166 |         margin: 1rem;
167 |     }
168 |
169 |     h1 {
170 |         font-size: 2rem;
171 |     }
172 | }
```

File: static/index.js — Core Client Form Validation RegEx Helpers & Countdown Utilities


```

1 | /**
2 |  * Event Entry & Pass Management System - Client Logic
3 |  * Handles real-time form validation, QR streaming, timer synchronization, and notifications.
4 |  */
5 |
6 | // Regular Expression Validation Patterns
7 | const REGEX_PATTERNS = {
8 |   name: /^[a-zA-Z\s.]{3,50}$/ ,
9 |   phone: /^[6-9]\d{9}$/ ,
10 |   duration: /^\\d+$/
11 | };
12 |
13 | /**
14 |  * Validates participant entry form inputs
15 |  * @param {string} name - Participant full name
16 |  * @param {string} phone - 10-digit mobile number
17 |  * @param {number} duration - Session duration in minutes
18 |  * @returns {object} { isValid: boolean, message: string }
19 |  */
20 | function validateEntryForm(name, phone, duration) {
21 |   if (!REGEX_PATTERNS.name.test(name.trim())) {
22 |     return { isValid: false, message: "Please enter a valid full name (3-50 letters only)." };
23 |   }
24 |   if (!REGEX_PATTERNS.phone.test(phone.trim())) {
25 |     return { isValid: false, message: "Please enter a valid 10-digit Indian mobile number." };
26 |   }
27 |   if (isNaN(duration) || parseInt(duration) < 1) {
28 |     return { isValid: false, message: "Duration must be at least 1 minute." };
29 |   }
30 |   return { isValid: true, message: "Validation successful." };
31 | }
32 |
33 | /**
34 |  * Formats duration in minutes to human-readable string
35 |  * @param {number} minutes
36 |  * @returns {string} e.g. "2h 30m"
37 |  */
38 | function formatDuration(minutes) {
39 |   const hrs = Math.floor(minutes / 60);
40 |   const mins = minutes % 60;
41 |
42 |   if (hrs > 0) {
43 |     return mins > 0 ? `${hrs}h ${mins}m` : `${hrs}h`;
44 |   }
45 |   return `${mins}m`;
46 | }
47 |
48 | /**
49 |  * Calculates remaining countdown time from target timestamp
50 |  * @param {string|Date} endTimeIso
51 |  * @returns {object} { totalSeconds, isExpired, formatted }
52 |  */
53 | function calculateCountdown(endTimeIso) {
54 |   const target = new Date(endTimeIso).getTime();
55 |   const now = new Date().getTime();
56 |   const diff = Math.max(0, Math.floor((target - now) / 1000));
57 |
58 |   const minutes = Math.floor(diff / 60);
59 |   const seconds = diff % 60;
60 |
61 |   return {
62 |     totalSeconds: diff,
63 |     isExpired: diff <= 0,
64 |     formatted: `${String(minutes).padStart(2, '0')}:${String(seconds).padStart(2, '0')}`
65 |   };

```